TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the End Corrections of a Metre Bridge

The bridge wire does not start exactly at the scale's zero — the copper end-strips add a hidden length at each end. Interchanging two known resistances exposes both corrections, α and β.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 2 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine the end corrections α (left end) and β (right end) of a metre bridge. In the ideal metre bridge, an unknown X in the right gap balances a known S in the left gap at length l from the left end, giving X = S(100 − l)/l. In the real instrument the wire is soldered to thick copper strips, and the junctions, the strips themselves and the contact resistances behave as if the wire were a little longer at each end: the effective left segment is (l + α) and the right is (100 − l + β), where α and β are small lengths (typically a few tenths of a centimetre) called the **end corrections**. The true balance equation is therefore

$$S / X = (l + \alpha) / (100 - l + \beta)$$

Two unknowns need two measurements. Take two known resistances P and Q: balance with P in the left gap and Q in the right at l_1; then **interchange** them and balance again at l_2. The two equations

$$P/Q = (l_1 + \alpha)/(100 - l_1 + \beta) \quad \text{and} \quad Q/P = (l_2 + \alpha)/(100 - l_2 + \beta)$$

contain only α and β as unknowns, and solving them gives both corrections. Once known, they correct every later measurement made on that bridge.

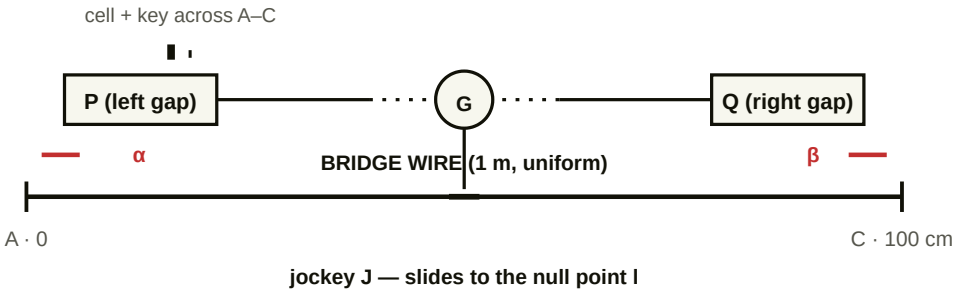


Figure 1 — The metre bridge. The cell drives current along the wire; the galvanometer between the middle terminal and the jockey reads zero at balance. The end corrections α and β (red) extend the wire invisibly into the copper strips.

2. Apparatus and procedure

- **Apparatus:** metre bridge, two known resistances P and Q of comparable size (e.g. 2 Ω and 3 Ω), cell, galvanometer, jockey, key, connecting wires.
- Wire P into the left gap and Q into the right gap, with the cell across the wire ends and the galvanometer from the centre terminal to the jockey.
- Press the key and slide the jockey gently to the null point (galvanometer zero, no kick on make or break). Record l_1 to 0.1 cm, approaching the null from both sides to bracket it.
- Interchange P and Q. Find the new null point l_2 the same way.
- Substitute l_1, l_2, P, Q into the two balance equations and solve the simultaneous pair for α and β.
- Repeat with a second pair of resistances and average α and β.

3. Worked calculation

Data: P = 2 Ω, Q = 3 Ω; null at l_1 = 39.9 cm (ideal, uncorrected: 40.0). Interchanged: l_2 = 60.0 cm.

$$2/3 = (39.9 + \alpha)/(60.1 + \beta) \rightarrow 3\alpha - 2\beta = 0.5$$

$$3/2 = (60.0 + \alpha)/(40.0 + \beta) \rightarrow 2\alpha - 3\beta = 0$$

Solving the pair: **α = 0.30 cm, β = 0.20 cm**. Thereafter every balance on this bridge is corrected — e.g. an unknown balancing S = 5 Ω at l = 33.0 cm is really $X = S(100 - l + \beta)/(l + \alpha) = 5 \times 67.2/33.3 \approx 10.09$ Ω, not the 10.15 Ω the naked scale gives.

4. Precautions

- Keep the current brief — a warm wire changes resistance and the corrections with it. Read quickly, then open the key.
- Press the jockey vertically without dragging it along the wire (dragging wears the wire non-uniform and scratches off the calibration).
- All terminals and gap contacts must be clean and tight — stray contact resistance masquerades as end correction.
- Choose P and Q comparable so both null points fall in the middle region (30–70 cm), where the bridge is most sensitive and the corrections weigh most evenly.

Practice (viva preparation)

1. What physical causes make up the end correction?
2. Why does one balance reading not suffice to find α and β, and why does interchanging P and Q provide the second equation?
3. A bridge has α = 0.4 cm, β = 0.1 cm. A known 4 Ω in the left gap balances an unknown at l = 44.0 cm. Find the unknown with and without correction.
4. Why should the null point be kept near the middle of the wire?
5. Two resistances are exactly equal. Where should the null fall with α = 0.3, β = 0.2, and why not at 50.0?

Answer key

1. The wire's soldered joints to the copper strips, the resistance of the strips themselves between the gap terminals and the wire ends, and contact resistance at the terminals — together they act like a small extra length of wire at each end.
2. One reading gives one equation in two unknowns (α, β). Interchanging swaps the roles of the gaps while α and β stay attached to the bridge's own ends, producing a genuinely independent second equation; the pair then determines both.
3. Corrected: $X = S(100 - l + \beta)/(l + \alpha) = 4 \times 56.1/44.4 \approx 5.05\ \Omega$. Uncorrected: $4 \times 56/44 \approx 5.09\ \Omega$ — the corrections move the answer by about 0.8 % here.
4. Sensitivity of the bridge is greatest near the middle (the galvanometer current changes fastest with jockey movement), the wire is most uniform there, and percentage errors in l enter both segments most evenly.
5. Balance needs $(l + \alpha) = (100 - l + \beta)$: $l = (100 + \beta - \alpha)/2 = (100 - 0.1)/2 = 49.95$ cm — a hair left of centre, because the left end hides slightly more extra length than the right.